The flux landscape has exactly one exact structure: the D3 tadpole

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Abstract

We prove that the D3 tadpole $N_{\text{flux}} = F^\top JH$ is a *complete* invariant of the type IIB flux under the full duality group $\mathrm{Sp}(b_3) \times \mathrm{SL}(2,\mathbb{Z})$: two fluxes are dual-equivalent if and only if they have the same tadpole. The proof is two lemmas. First (Mitchell's Uniqueness Theorem, upgraded by the First Fundamental Theorem of invariant theory, Weyl 1939), the ring of duality-invariant polynomials in the flux pair is generated by the tadpole alone — there is no invariant of any degree other than N_{flux} and its powers. Second, the duality group acts transitively on each non-zero tadpole level set (the orbit tangent space has rank $2b_3 - 1$, equal to the level set's codimension-1 dimension). Together they make N_{flux} a complete invariant. Consequence — a *no-fingerprint no-go*: no exact, frame-independent quantity can distinguish two vacua that share a tadpole, and since the tadpole is bounded by the orientifold charge ($N_{\text{flux}} \le L$), a landscape search can only *eliminate* (by the constraint), never *select*. The exact content of a theory of everything is thereby constrained to two classes — consistency constraints and topological labels — and nothing else. The statement is falsifiable: a single moduli-independent, duality-invariant flux quantity that is not a function of $F^\top JH$ breaks it. All claims are verified symbolically (sympy, zero tolerance) and reproduced by `scripts/validate_all_theorems.py`.

1 Introduction

The constraint-first program measured, on 174 supersymmetric IIB flux vacua, that the low-energy observables of a flux vacuum form a mostly-unstructured cloud: the string coupling, the gauge couplings, the mass-spectrum shape, the axions, the dilaton and the cosmological constant are statistically uniform in flux, while exactly one quantity — the D3 tadpole — is exactly structured [**?**]. That was a measurement. This paper upgrades its central claim — *"the landscape's only exact structure is what it forbids"* — from an empirical pattern to a theorem, by identifying the flux pair as the $k = 2$ case of a classical result in invariant theory.

2 Setup

Let X be a Calabi–Yau threefold with middle cohomology $H^3(X,\mathbb{Z}) \simeq \mathbb{Z}^{b_3}$, and let the two NS–NS/RR three-form fluxes be $F, H \in H^3(X,\mathbb{Z})$. The intersection form is the antisymmetric $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ in a symplectic basis, and the D3 tadpole is the symplectic pairing

$$N_{\text{flux}} \;=\; F^\top JH \;=\; \sum_{i=0}^{b_3/2-1} \big(F_i H_{i+b_3/2} - F_{i+b_3/2} H_i\big). \tag{1}$$

The duality group is $\mathrm{Sp}(b_3,\mathbb{Z})$ (the monodromy group acting on H^3, hence on F and H separately) together with S-duality $\mathrm{SL}(2,\mathbb{Z})$ acting as $(F,H) \mapsto (aF + bH, cF + dH)$, $ad - bc = 1$. The tadpole is invariant under both: under $\mathrm{Sp}(b_3)$ it is the symplectic pairing, and under $\mathrm{SL}(2,\mathbb{Z})$ one computes $N_{\text{flux}} \mapsto (ad - bc) N_{\text{flux}}$ because $F^\top JF = H^\top JH = 0$ and $H^\top JF = -F^\top JH$.

3 Mitchell's Uniqueness Theorem

Theorem 1 (Mitchell)**.** *The D3 tadpole $F^\top JH$ is the unique non-trivial bilinear form in the flux (F, H) that is invariant under $\mathrm{Sp}(b_3, \mathbb{Z})$, for any b_3.*

The candidates $F^\top F$, $H^\top H$, $F^\top H$ are gauge-dependent (they change under a symplectic transformation), while $F^\top JF = H^\top JH = 0$ identically by antisymmetry. The remaining pairing is $F^\top JH$. This is verified symbolically (6/6 checks, `scripts/validate_mitchell_uniqueness.py`).

4 The Constraint-First Theorem: the full invariant ring

Mitchell's theorem is the degree-2 slice. The whole curve is classical.

Theorem 2 (Constraint-First, = First Fundamental Theorem, Weyl 1939)**.** *Every $\mathrm{Sp}(b_3)$-invariant polynomial in the flux pair (F, H) is a polynomial in the tadpole:*

$$\mathbb{C}\big[(\mathbb{C}^{b_3})^2\big]^{\mathrm{Sp}(b_3)} \;=\; \mathbb{C}\big[F^\top JH\big]. \tag{2}$$

This is Weyl's First Fundamental Theorem of invariant theory for the symplectic group specialized to $k = 2$ copies of the defining representation; the flux pair $(F, H) \in \mathbb{C}^{b_3} \oplus \mathbb{C}^{b_3}$ is precisely that $k = 2$ case. It says the tadpole *generates* the invariant ring: the only invariants are $1,\ F^\top JH,\ (F^\top JH)^2,\ \ldots$ We verify the completeness direction computationally at every degree $d = 1, \ldots, 5$ by annihilating a torus-reduced ansatz with the full $\mathfrak{sp}(b_3)$ algebra and computing the nullity exactly; the invariant-space dimensions are

$$\dim = 0,\ 1,\ 0,\ 1,\ 0 \qquad (d = 1, 2, 3, 4, 5), \tag{3}$$

the signature of a ring generated by a single degree-2 element (`scripts/validate_constraint_first.py`).

Remark 1. *The honest attribution is that this is* not new mathematics*: it is Weyl's* 1939 *theorem. The contribution here is the* identification *— that the flux pair is the $k = 2$ FFT case, so the "only exact structure is the constraint" is a corollary of classical invariant theory rather than an empirical conjecture.*

5 The No-Fingerprint No-Go Theorem

Theorem 3 (No-Fingerprint No-Go)**.** *For $N_{\mathrm{flux}} \neq 0$, two flux pairs are dual-equivalent under $\mathrm{Sp}(b_3)$ if and only if they have the same tadpole. Equivalently, N_{flux} is a* complete *invariant.*

Sketch. Necessity is the Constraint-First Theorem: an invariant is a function of N_{flux}. Sufficiency is transitivity. The orbit tangent space at a pair is the image of the map $\mathfrak{sp}(b_3) \to \mathbb{R}^{2b_3}$, $X \mapsto (XF, XH)$, whose rank is $2b_3 - 1$; this equals the codimension-1 dimension of the level set $\{F^\top JH = N\}$. The orbit is open in the level set (equal tangent spaces), closed (an algebraic group orbit), and the level set is connected for $N \neq 0$, so the orbit *is* the level set. □

The rank is computed exactly (integer rref) at three generic pairs and at the canonical pair (e_0, e_3) itself, giving $2b_3 - 1 = 11$ for $b_3 = 6$ (`scripts/validate_no_fingerprint.py`).

Corollary 1 (Search can only eliminate, never select)**.** *No exact, frame-independent quantity can distinguish two vacua that share a tadpole. The only exact selector is N_{flux} itself, and it is* bounded *($N_{\mathrm{flux}} \leq L$ by the orientifold charge). Therefore a landscape search can only* eliminate *by the constraint — it can never* select *a special vacuum, because there is no invariant left to select with.*

6 Full-duality completion

Theorem 4. *The invariant ring of the* full *duality group* $\mathrm{Sp}(b_3) \times \mathrm{SL}(2,\mathbb{Z})$ *is also* $\mathbb{C}[F^\top JH]$*, so the no-go holds for the complete duality group, not just the monodromy subgroup.*

By the Constraint-First Theorem the $\mathrm{Sp}(b_3)$-invariants are $\mathbb{C}[N_{\text{flux}}]$; the generator N_{flux} is $\mathrm{SL}(2,\mathbb{Z})$-invariant (verified symbolically for the generators $S : (F,H) \mapsto (-H,F)$ and $T : (F,H) \mapsto (F, F+H)$, and for a general SL(2) element $N_{\text{flux}} \mapsto (ad-bc)N_{\text{flux}}$), so $\mathrm{SL}(2,\mathbb{Z})$-invariance removes nothing. The full invariant ring is exactly $\mathbb{C}[N_{\text{flux}}]$ (`scripts/validate_full_duality.py`).

7 The Complete Exact Structure Theorem

Theorem 5 (Complete Exact Structure)**.** *For a* CY_3 *orientifold flux compactification, every exact (moduli-independent, duality-invariant) observable is either (i) the D3 tadpole* N_{flux} *— the unique flux observable — or (ii) a topological invariant of the Calabi–Yau* $(h^{1,1}, h^{2,1}, c_2, \kappa)$*. There is nothing else.*

In particular W_0, g_s, the masses, the saxion values, the gauge couplings and the cosmological constant are *not* exact observables: they are moduli- or flux-dependent, and duality-frame-dependent, hence exactly the quantities a theory of everything does *not* get to predict.

Corollary 2 (Hard ceiling on the verification deficit)**.** *The flux sector contributes exactly* one *observable to the Mitchell Verification Deficit* m*. By the no-go there is provably no second flux observable to find, so* m *can never grow from the flux sector; future growth must come from the other sectors (D-brane charges, the Standard Model), which is the* $m = 0$ *frontier.*

Remark 2 (Falsifiability)**.** *The whole tower stakes itself on one checkable claim:* no moduli-independent, duality-invariant flux quantity exists that is not a function of $F^\top JH$*. A single counterexample falsifies it. That is what makes this a theorem rather than a slogan.*

8 Conclusion

The constraint-first conjecture is now a theorem. The flux sector's only exact structure is the D3 tadpole — a *constraint*, not a selector — and the theory of everything predicts nothing exactly except what it forbids (consistency constraints) and what it is handed (topology). Search is provably dead; the only navigable move is elimination by verified constraint.

Scope, named. The invariant-ring claim is proven for *polynomial* invariants (the FFT); smooth non-polynomial invariants are not covered by the symbolic proofs, though nothing suggests they exist. Global transitivity rests on two classical facts (the orbit is closed; the $N \neq 0$ level set is connected), stated as such; their machine-checked (Lean) formalization is deferred to hardware. The topology list $h^{1,1}, h^{2,1}, c_2, \kappa$ is the standard complete topological data of a CY_3, taken as established mathematics.

References

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